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# Question

- ① Two circles  $x^2 + y^2 = 6$  and  $x^2 + y^2 - 6x + 8 = 0$  are given. Then the equation of the circle through their points of intersection and the point  $(1, 1)$  is (1980)
- ①  $x^2 + y^2 - 6x + 4 = 0$
  - ②  $x^2 + y^2 - 3x + 1 = 0$
  - ③  $x^2 + y^2 - 4y + 2 = 0$
  - ④ none of these

# Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,  
equation circles are

$$\mathbf{x}^T \mathbf{x} - 6 = 0 \quad (1)$$

$$\mathbf{x}^T \mathbf{x} + 2 \begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{x} + 8 = 0 \quad (2)$$

substitute 1.1 in 1.2

$$\begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{x} = -7 \quad (3)$$

$$x = \frac{7}{3} \quad (4)$$

substitute 1.4 in 1.1

$$y = \frac{\sqrt{5}}{3}, \frac{-\sqrt{5}}{3} \quad (5)$$

let equation of required circle is

$$\mathbf{X}^T \mathbf{X} + 2\mathbf{u}^T \mathbf{X} + f = 0 \quad (6)$$

$$\mathbf{u}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{-f}{2} - 1 \quad (7)$$

$$\mathbf{u}^T \begin{pmatrix} \frac{7}{3} \\ \frac{\sqrt{5}}{3} \end{pmatrix} = \frac{-f}{2} - 3 \quad (8)$$

$$\mathbf{u}^T \begin{pmatrix} \frac{7}{3} \\ -\frac{\sqrt{5}}{3} \end{pmatrix} = \frac{-f}{2} - 3 \quad (9)$$

$$\begin{pmatrix} 1 & 1 \\ \frac{7}{3} & \frac{\sqrt{5}}{3} \\ \frac{7}{3} & -\frac{\sqrt{5}}{3} \end{pmatrix} \mathbf{X} = \begin{pmatrix} \frac{-f}{2} - 1 \\ \frac{-f}{2} - 3 \\ \frac{-f}{2} - 3 \end{pmatrix} \quad (10)$$

# Theoretical Solution

$$\text{rank} \begin{pmatrix} 1 & 1 & \frac{-f}{2} - 1 \\ \frac{7}{3} & \frac{\sqrt{5}}{3} & \frac{-f}{2} - 3 \\ \frac{7}{3} & \frac{-\sqrt{5}}{3} & \frac{-f}{2} - 3 \end{pmatrix} = 2 \quad (11)$$

$$f = 1 \quad (12)$$

$$\mathbf{u} = \begin{pmatrix} \frac{-3}{2} \\ 0 \end{pmatrix} \quad (13)$$

the required circle equation is

$$x^2 + y^2 - 3x + 1 = 0 \quad (14)$$

# Plot

